LEARNER GUIDE

Percentages as multipliers

Percentages, Ratio, Growth and Bounds · Lesson 3. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A **percentage** is a number out of 100: 35% = 0.35, so 35% of £80 = 0.35 × £80 = £28. The whole amount is 100%, a **multiplier** of 1. An increase of 15% leaves 115% of the amount, so multiply by 1.15. A decrease of 20% leaves 80%, so multiply by 0.8. To write one number as a percentage of another, divide the part by the whole and multiply by 100.

What you are learning to notice

- 100% is a multiplier of 1. Above 1 is an increase; below 1 is a decrease.
- An increase of 5% is × 1.05, not × 1.5.
- × 0.2 finds 20% of an amount; × 0.8 decreases it by 20%.
- One multiplier replaces two steps (find the change, then add or subtract it).
- Up 10% then down 10% is × 0.99: not back to the start.

Moving through it, one step at a time

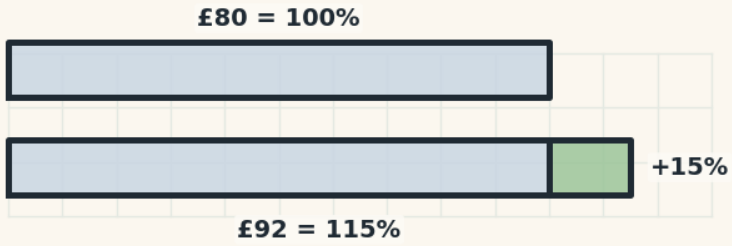
Write the multiplier: 100% plus the increase, or 100% minus the decrease, then divide by 100. Multiply the original amount by it. For a percentage of another number: part ÷ whole × 100. Estimate first, then use a calculator if you like.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Increase £80 by 15%

A jacket costs £80. Its price rises by 15%. What is the new price?

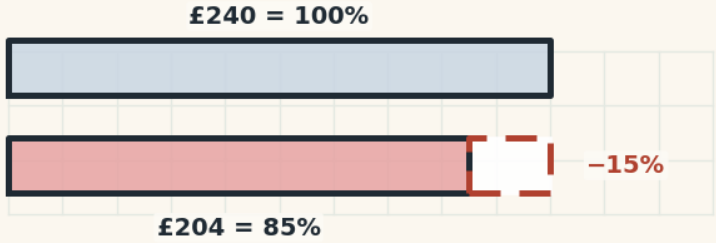


1. The new price is 100% + 15%	115% = 1.15
2. Multiply once	£80 × 1.15 = £92
3. Check with two steps	15% of £80 = £12; £80 + £12 = £92

The new price is £92.

Example 2 — Decrease £240 by 15%

A £240 phone is reduced by 15% in a sale. What is the sale price?



1. What is left: 100% − 15%	85% = 0.85
2. Multiply once	£240 × 0.85 = £204

The sale price is £204.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (fractions, decimals and percentages as one number, and multiplying by a decimal), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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